

3(a)	any four from: 1 protons / H^+ ions, pumped, from <u>guard cells</u> / into <u>guard cell</u> wall ; 2 <i>ref. to</i> negative potential (inside cell) or Cl^- ions enter (cell) ; 3 (voltage-gated) K^+ channels open or K^+ ions enter (guard) cells ; 4 water enters cell, by osmosis / down water potential gradient ; 5 (guard) cells, become turgid / swell / increase in volume ;	4
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Question	Answer	Marks
3(b)(i)	any two from: 1 as abscisic acid concentration increases stomatal conductance decreases or negative correlation / inverse correlation / inversely proportional ; 2 two sets of x y figures to support mp 1 with units (at least once) ; 3 steep decrease in conductance to 100–150 ABA ; 4 no / little, effect on conductance above 250–425 ABA ;	2
3(b)(ii)	any four from: 1 ABA causes stomata to close ; 2 ABA binds to receptor on <u>guard</u> cell ; 3 Ca²⁺ ions, enter (guard) cells / act as 2nd messenger ; 4 K⁺ ions leave (guard) cells ; 5 water leaves so cells become, flaccid / less turgid ; 6 AVP ;	4
3(c)	DNA / promoter ;	1

Question	Answer	Marks
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